Estimating parameters

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Agenda and reminders

- HW 1 due Friday
- Project part 1 released later this week (choosing a dataset + EDA)
- This week: parameter estimation

Goal

Logistic regression model:

$$Y_i \sim Bernoulli(\pi_i) \quad \log\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_0 + \beta_1 X_i$$

> Given data $(X_1, Y_1), (X_2, Y_2), \ldots, (X_n, Y_n)$, how do I estimate β_0 and β_1?

Motivating example

$=1$ $=0$

$Y_i =$ result of flipping a coin (Heads or Tails)

Let's make a model:

- **Step 1:** Distribution of the response

probability of heads

$$Y_i \sim Bernoulli(\pi)$$

- **Step 2:** Construct a model for the parameters

$$\pi = ??$$

Motivating example

$Y_i =$ result of flipping a coin (Heads or Tails)

Suppose your friend estimates that the probability of heads is 0.9

- $Y_i \sim Bernoulli(\pi)$
- $\widehat{\pi} = 0.9$

> How can we assess whether this estimate $\widehat{\pi}$ is reasonable?

Flip the coin (many times, ideally)

(multiplication rule for independent events)

Motivating example

Suppose we flip the coin 5 times, and observe

$$y_1, \ldots, y_5 = T, T, T, T, H$$

> What is the probability of (i.e., how *likely* is) getting this string of flips if $\pi = 0.9$?

$$P(T, T, T, T, H \mid \pi = 0.9)$$

$$= P(T \mid \pi = 0.9) \cdot P(T)\, P(T)\, P(T)\, P(H \mid \pi = 0.9)$$

$$= (1 - P(H \mid \pi = 0.9))^4 \, P(H \mid \pi = 0.9)$$

$$= (1 - \pi)^4 \, \pi$$

$$= (0.1)^4 \, 0.9 \qquad \text{for } \pi = 0.9$$

$$= 0.00009$$

Likelihood

Definition: The *likelihood* $L(Model) = P(Data|Model)$ of a model is the probability of the observed data, given that we assume a certain model and certain values for the parameters that define that model.

- Model: $Y_i \sim Bernoulli(\pi)$, and $\widehat{\pi} = 0.9$
- Data: $y_1, \ldots, y_5 = T, T, T, T, H$
- Likelihood: $L(\widehat{\pi}) = P(y_1, \ldots y_5 | \pi = 0.9) = 0.00009$

Class Activity, Part I

Work on the activity (handout) with a neighbor, then we will discuss as a class

$$L(0.2) = P(Y, N, N, Y, N \mid \pi = 0.2)$$
$$= (0.2)(0.8)(0.8)(0.2)(0.8) = (0.2)^2(0.8)^3 = 0.020$$

$$L(0.3) = P(Y, N, N, Y, N \mid \pi = 0.3)$$
$$= (0.3)^2(0.7)^3 = 0.031$$

Class Activity

$$L(0.2) = P(y_1, \ldots, y_5 | \pi = 0.2) \\ = (0.2)(0.8)(0.8)(0.2)(0.8) = 0.020$$

$$L(0.3) = P(y_1, \ldots, y_5 | \pi = 0.3) \\ = (0.3)(0.7)(0.7)(0.3)(0.7) = 0.031$$

> Which value, 0.2 or 0.3, seems more reasonable?

higher likelihood!

Class Activity

> Which value of $\widehat{\pi}$ in the table would you pick?

$\hat{\pi} = 0.4$ has the highest likelihood (of the values considered in the table)

Also: observed proportion of $Y = \frac{2}{5} = 0.4$

Maximum likelihood

Maximum likelihood principle: estimate the parameters to be the values that maximize the likelihood

$\widehat{\pi}$	**Likelihood**
0.30	0.031
0.35	0.033
0.40	0.036
0.45	0.033

Maximum likelihood estimate: $\widehat{\pi} = 0.4$

Maximum likelihood

Maximum likelihood principle: estimate the parameters to be the values that maximize the likelihood

Steps for maximum likelihood estimation:

- *Likelihood*: For each potential value of the parameter, compute the likelihood of the observed data
- *Maximize*: Find the parameter value that gives the largest likelihood

Maximum likelihood for logistic regression

$$\log\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_0 + \beta_1 \, Size_i \qquad \pi_i = \frac{\exp\{\overset{-2}{\beta_0} + \overset{0.5}{\beta_1} \, Size_i\}}{1 + \exp\{\underset{-2}{\beta_0} + \underset{0.5}{\beta_1} \, Size_i\}}$$

Observed data:

Tumor cancerous	Yes	No	No	Yes	No
Size of tumor (cm)	6	1	0.5	4	1.2
	$\hat{\pi}_1$	$\hat{\pi}_2$	$\hat{\pi}_3$	$\hat{\pi}_4$	$\hat{\pi}_5$

> Suppose $\beta_0 = -2, \ \beta_1 = 0.5$. How would I compute the likelihood?

1) Plug in $\beta_0 = -2$, $\beta_1 = 0.5$, and tumor size

$\Rightarrow$ give $\hat{\pi}_i$ for each observation

$(\hat{\pi}_1, \hat{\pi}_2, \ldots, \hat{\pi}_5$ if $\beta_0 = -2$, $\beta_1 = 0.5$ are the values)

2) $L(\beta_0 = -2, \beta_1 = 0.5) = P(Y|\hat{\pi}_1)\, P(N|\hat{\pi}_2)\, P(N|\hat{\pi}_3)\, P(Y|\hat{\pi}_4)\, P(N|\hat{\pi}_5)$

$= \hat{\pi}_1 (1-\hat{\pi}_2)(1-\hat{\pi}_3)\, \hat{\pi}_4 (1-\hat{\pi}_5)$

Class Activity, Part II

Work on the activity (handout) with a neighbor, then we will discuss as a class

Class Activity

$$\widehat{\pi}_i = \frac{\exp\{-2 + 0.5\ Size_i\}}{1 + \exp\{-2 + 0.5\ Size_i\}}$$

Tumor cancerous	Yes	No	No	Yes	No
Size of tumor (cm)	6	1	0.5	4	1.2
$\widehat{\pi}_i$	0.73	0.18	0.15	0.5	0.2

Likelihood = $(0.73)(1-0.18)(1-0.15)(0.5)(1-0.2)$

≈ 0.2

$L(\hat{\beta}_0 = -2, \hat{\beta}_1 = 0.5) \approx 0.2$

Maximum likelihood for logistic regression

Likelihood:

- For estimates $\widehat{\beta}_0$ and $\widehat{\beta}_1$, $\widehat{\pi}_i = \dfrac{\exp\{\widehat{\beta}_0 + \widehat{\beta}_1 X_i\}}{1 + \exp\{\widehat{\beta}_0 + \widehat{\beta}_1 X_i\}}$
- $L(\widehat{\beta}_0, \widehat{\beta}_1) = P(Y_1, \ldots, Y_n | \widehat{\beta}_0, \widehat{\beta}_1)$

Maximize:

- Choose $\widehat{\beta}_0, \widehat{\beta}_1$ to maximize $L(\widehat{\beta}_0, \widehat{\beta}_1)$

1) Calculus
(write down likelihood, differentiate)
↳ next time!
2) Do it all by hand
3) Do it on a computer

> So far, we only considered a few values for β_0 and β_1. How should we check other values, to make sure our estimates actually maximize likelihood?